

S^n , ~~S^1~~ , RP^n , $\langle \dots \rangle$
 $(n \geq 1)$
 $\downarrow$
 $\pi_1 = 0$

$M \cup N$ $\wedge \vee$ $\langle X, S \rangle / \langle \dots \rangle, \langle \dots \rangle$
 π_0, π_1
 $a \rightarrow b$

$\mathbb{Z} \times \mathbb{Z}$ $\langle a, b \mid aba = b \rangle$
 $ababab \dots$
 $abab^t = e$

$ab = ba \rightarrow$ group $\mathbb{Z} \times \mathbb{Z}$
 $\langle a, b \mid ab = ba \rangle$
 $\mathbb{Z} \times \mathbb{Z}$

$S^1 \vee S^2$
 $\pi_1 = \mathbb{Z} \times \mathbb{Z}$
 $\pi_2 = \mathbb{Z}$

$\pi_1(RP^n) = \mathbb{Z}$
 $(n \geq 2)$
 $\pi_1(M) \cong \pi_1(M \cup \dots)$

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